

Rajarshi Ray

Founder, SceneSenseAI · AI Engineer, IBM

New York, USA rajarshi5675@gmail.com +1 (631) 949-3136 rajarshi-ray29.github.io github.com/rajarshi-ray29
linkedin/rajarshi-ray

EDUCATION

M.S., Computer Science — Stony Brook University (SUNY), NY Aug 2024 – Dec 2025
GPA 3.9/4.0 · Coursework: Computer Vision, Natural Language Processing, AI, Data Science

B.E., Computer Science — Birla Institute of Technology and Science (BITS), Pilani 2020 – Jul 2024
GPA 8/10 · Coursework: AI, Data Science, Data Visualization · Teaching Assistant, Object-Oriented Programming

INDUSTRY EXPERIENCE

Staff Engineer, Artificial Intelligence — IBM, Poughkeepsie, NY Feb 2026 – Present

- Architecting a **patent-pending agentic AI system** generating **100K+ synthetic financial transactions per run** for large-scale workload testing on IBM z17 LinuxONE — fully on-prem to meet enterprise security requirements.
- Collaborating with **IBM Research** to benchmark agentic AI workflows for industrial robotics, evaluating **4 agent architectures across 4,000+ real-world asset-inspection scenarios**; targeting a submission to ICLR 2027.
- Running a locally deployed **AI playground on IBM GPU infrastructure**: 10 self-hosted LLMs serving 20–25 engineers for day-to-day experimentation and prototyping.

Software Engineer Intern — IBM, New York May 2025 – Feb 2026

- Deployed and ported **6 microservices** across x86 and s390x architectures using Podman and OpenShift Pipelines.
- Engineered a **JMeter-based pipeline** for automated workload testing, achieving a **98% success rate** in simulating real-world usage.
- Contributed to the **WatsonX Challenge** by designing an agentic AI framework for persona-driven synthetic transaction data generation.

ENTREPRENEURSHIP

Founder — SceneSenseAI 2026 – Present

- Building **personalized AI teaching assistants** that connect to lecture/course video libraries: learners ask questions in natural language and receive answers **cited with exact timestamps** into the source videos.
- Designed an **instructor-facing analytics engine** that surfaces learner confusion points and content gaps per module — before they appear in exam results or refund requests.
- Validating with online course creators and university instructors; founding pilot program in progress across the US.

RESEARCH EXPERIENCE

InspectionBench — Embodied Industrial Anomaly Verification — IBM × IBM Research 2026 – Present

- Building a **perception-grounded physical-AI evaluation benchmark**: 4 agent architectures across **4,000+ real-world asset-inspection scenarios**; manuscript in preparation targeting **ICLR 2027**.

NTIRE 2025 Image Super-Resolution ×4 Challenge — VAI-GM Lab, Stony Brook University

Jan 2025 – May 2025

- Led Team VAI-GM; designed the **Multi-step Dense Residual Connected Transformer (MDRCT)**, improving on the DRCT baseline with multi-step residual connections for better gradient flow and feature preservation.
- Achieved **30.86 dB PSNR / 0.8942 SSIM**, ranking **18th globally**; published in the CVPR 2025 NTIRE Workshop challenge report.

Grid-Based All-in-One Image Degradation & Restoration — VAI-GM Lab, Stony Brook University

May 2025 — Feb 2026

- Curated a dataset of **80,000+ grid images** by synthesizing rain, haze, noise, and snow degradations; designed a transformer-based model performing multi-degradation restoration with a single network.

3D Lung CT Synthesis with Guided Diffusion — VAI-GM Lab, advised by Prof. Klaus Mueller

Aug 2024 — May 2025

- Generated high-fidelity synthetic 3D lung CT images by encoding medical images into latent space with **VQ-VAE** and sampling with guided **diffusion models**.
- Built and maintained an **Orthanc-based DICOM server** (Docker) for scalable, secure medical-image storage, retrieval, and analysis.

PUBLICATIONS

Chen Z., **Ray R.**, Tomar S.S., Mueller K., et al. (2025). "NTIRE 2025 Image Super-Resolution (×4) Challenge Report." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), NTIRE Workshop*. [arXiv:2504.14582](https://arxiv.org/abs/2504.14582)

In preparation: "InspectionBench: Embodied Industrial Anomaly Verification with Perception-Grounded Physical AI Evaluation" (with IBM Research) — targeting ICLR 2027.

SKILLS

AI / ML: PyTorch, TensorFlow/Keras, Diffusion Models, VAEs, GANs, Transformers, LLMs, Agentic AI, RAG, Model Quantization, Hugging Face, OpenCV, YOLO

Systems & Cloud: Docker, Podman, OpenShift/Tekton, Kubernetes, Linux (x86/s390x), JMeter, GitOps, DICOM/Orthanc

Languages: Python, C/C++, Java, Kotlin, SQL, Bash

HIGHLIGHTS

- Patent-pending** agentic AI system for synthetic financial data (IBM)
- Ranked 18th globally** — NTIRE 2025 Super-Resolution Challenge, CVPR 2025
- IBM WatsonX Challenge** — agentic synthetic-data framework
- Founder** — SceneSenseAI, AI teaching assistants for video course libraries